

```

int z;
keyldy(); //
key debounce
writecmd(0xC0); //
select second line on LCD
if(y<14)
if not max pulse width
{
x--;
y++; //
increase it convert it in to
z=y*5*0x30; //
1 to 10 scale and ASCII
writestr("speed: "); // dia-
play speed on LCD
writedata(z); //
writestr(" "); //
}
else if(y==14)
writestr("min speed: 9 "); //
if max speed display message
}
void incspeed() //
increase speed
{
int w;
keyldy();
writecmd(0xC0); // key de-
bounce
if(y>6) //
if not minimum width
{
x++;
y--; // decrease
it
w=y*5*0x30; // do same
as above
writestr("speed: ");
writedata(w); //
writestr(" "); //
}
else if(y==6)
writestr("max speed: 1 "); //
if min speed display message
}
void incptime() //
increase time
{
int p;
keyldy(); //
key debounce delay
writecmd(0xC0);
if(t<180) //
if not max time
{
t+=20; //
increase it by 1 sec
p=t/20;
p=p*0x30; //
convert it in to ASCII
writestr("time: "); //
// display it
writedata(p);
writestr(" "); //
}
else if(t==180)
writestr("max time: 9 sec"); //
// if max time display message
}
void decptime() //
decrease time
{
int q;
keyldy(); //
key debounce delay
writecmd(0xC0);
if(t>20) //
if not min time
{
t-=20; //
decrease it
q=t/20;
q=q*0x30; //
do same as above
writestr("time: ");
writedata(q);
}
}

writestr(" sec
");
}
else if(t==20)
writestr("min time: 1 sec"); //
// if min time display message
}
void keyldy() //
key debounce delay
{
int a,b;
for(a=0;a<50;a++)
for(b=0;b<1000;b++);
}
void time(unsigned int c) // change
time in seconds
{
int k;
TL1 = 0xAF;
// use timer 1
TH1 = 0x3C;
for(b=0;b<1000;b++);
}
void time(unsigned int c) // change
time in seconds
{
int k;
TL1 = 0xAF;
// use timer 1
TH1 = 0x3C;
for(b=0;b<1000;b++);
}
start timer
for(k=0;k<c;k++)
rotate loop in multiples of 20
{
while(TF1==0); //
wait till timer overflow
TF1 = 0; //
reset the flag
TL1 = 0xAF;
// reload it
TH1 = 0x3C;
}
TR1 = 0; //
stop timer
void delay(unsigned int cl) // change
time in micro seconds
{
int a;
TH0=0x9B; //
select timer 0
TL0=0x9B; //
to generate 100 micro second delay
TR0=1; //
start timer
for(a=0;a<c1;a++) //
rotate loop between 5 to 15
{
while(TF0==0); //
wait until timer overflow
TF0=0; //
reset the flag
TR0=0; //
stop timer
}
void main()
{
TMOD=0x12; // timer1 in 16
bit, timer 0 in 8 bit auto reload mode
P2=0xE0; //
LEDs off, relays OFF
P0=0x00; //
P0, P3 output ports
P3=0x00;
writecmd(0x3C); //
initialize LCD
writecmd(0x0E);
writecmd(0x01); //
writecmd(0x84); //
display message
writestr("DC Motor"); // DC motor
controller in
writecmd(0xC3); //
center of LCD
writestr("Controller"); //
agin:P1=0xFF; //
P1 as input port
while(P1==0xFF); //
wait until any key press
loop:switch(P1)
{
case 0xFE:
// for first key
keyldy(); //
key debounce
writecmd(0x01);
}
}

writestr("motor start");
time(50);
// wait for 2.5 sec
writecmd(0x80);
writestr("mode:continuous
"); // display current mode and speed
writecmd(0xC0);
writestr("speed: 5 "); //
led1=1; //
Run LED ON
led2=0; //
stop LED OFF
led1=1; //
clockwise direction ON
led4=0; //
anticlockwise direction OFF
start(); //
start rotating motor
break;
case 0xFD:
// for second key
keyldy(); //
key debounce
r=0; //
run flag reset
writecmd(0x01);
writestr("motor stop"); // dis-
play message
led1=0; //
Run OFF
led2=1; //
stop LED ON
led3=0; //
clockwise direction OFF
led4=0; //
anticlockwise direction OFF
low logic to PWM pin
RL1=1; //
relay1 off
break;
case 0xFB:
// for third key
mode(); //
select mode
if(r==1) start(); //
jump to start if run flag is set
break;
case 0xF7:
// for fourth key
direction(); //
change direction
if(r==1) start(); // jump to
start if run flag is set
break;
case 0xEF:
// for fifth key
incspeed(); //
increase speed
if(r==1) start(); // jump to
start if run flag is set
break;
case 0xDF:
// for sixth key
decspeed(); //
decrease speed
if(r==1) start(); // jump to
start if run flag is set
break;
case 0xBF:
// for seventh key
incptime(); //
increase time
if(r==1) start(); // jump to
start if run flag is set
break;
case 0x7F:
// for eighth key
decptime(); //
decrease time
if(r==1) start(); // jump to
start if run flag is set
break;
}
if(r==1) goto loop; //
if run flag is set jump of key detect
else goto agin; //
if not jump to again
}

```